Nmap Port Scan & Packet Analysis Lab

Overview

This lab demonstrates basic network reconnaissance and packet analysis using Nmap, Wireshark, and TShark inside an isolated virtual lab environment.
The objective of the project was to observe how TCP SYN-based port scanning behaves at the packet level and how such traffic appears during protocol analysis.

The lab was performed using two Kali Linux virtual machines configured on an internal virtual network.

Lab Environment

Component	Details
Attacker Machine	Kali Linux VM
Victim Machine	Kali Linux VM
Network Type	Internal Virtual Network
Hypervisor	VirtualBox
Packet Analyzer	Wireshark / TShark
Scanning Tool	Nmap

Environment Configuration

Network Setup

- Virtual machines configured using Internal Network
- Static IP addresses manually assigned
- Both systems placed in the same subnet

Example:

Victim IP : 192.168.11.11
Attacker IP: 192.168.11.12

Objective

The primary objective of this lab was to:

- Understand how TCP SYN scans operate
- Capture and inspect scan traffic using Wireshark
- Observe SYN, SYN-ACK, and RST packet behavior
- Analyze protocol hierarchy statistics
- Generate PCAP files for protocol-level analysis
- Practice packet inspection using Wireshark filters

Port Scan Activity

Nmap SYN Scan

The following command was executed from the attacker machine:

nmap -sS 192.168.11.11 --top-ports 100

Purpose

- Perform a TCP SYN stealth scan
- Scan the top 100 commonly used ports
- Identify open, closed, or filtered ports

Packet Capture

Wireshark Capture

Wireshark was launched on the victim machine to capture incoming scan traffic.

Observed Traffic

- TCP SYN packets
- TCP RST/ACK responses
- Port probing behavior
- Retransmissions
- Protocol distribution

Wireshark Filters Used

SYN Packet Detection

tcp.flags.syn == 1 && tcp.flags.ack == 0

Used to identify TCP SYN packets generated by the scan.

RST/ACK Packet Observation

tcp.flags.reset == 1

Used to observe reset packets returned from closed ports.

TCP Conversation Observation

tcp

Used for general TCP traffic inspection.

Protocol Hierarchy Analysis

Using:

Statistics → Protocol Hierarchy

The following protocols were observed:

- ARP
- IPv4
- TCP

- UDP
- DHCP
- ICMPv6

Observation

A significant portion of captured traffic consisted of:

- ARP communication
- TCP SYN packets
- TCP RST responses

This indicates active port probing and normal Layer-2 communication within the isolated network.

Technical Observations

TCP SYN Behavior

- SYN packets were sent from the attacker VM to multiple destination ports.
- The victim system responded depending on port state.

Closed Ports

Most scanned ports returned:

RST, ACK

indicating that the ports were closed.

Packet-Level Analysis

The scan generated:

- Multiple TCP connection attempts
- Short-lived TCP sessions
- Sequential port targeting behavior

Wireshark clearly displayed:

- Source IP
- Destination IP
- Destination ports
- TCP flags
- Sequence and acknowledgment values

TShark Capture

Traffic was also captured using TShark CLI.

Command Used

tshark -i eth0 -w nmap.pcap

Purpose

- Capture packets directly from terminal
- Save output as PCAP file
- Practice CLI-based packet analysis

Files Generated

File Type	Description
.pcap / .pcapng	Captured scan traffic
Screenshots	Wireshark observations
Notes	Technical observations
Filters	Protocol analysis filters

Tools Used

- Nmap
- Wireshark
- TShark
- Kali Linux
- VirtualBox

Learning Outcomes

Through this lab, the following concepts were explored:

- TCP SYN scanning
- Port state identification
- Packet capture techniques
- Wireshark filtering
- TCP flag analysis
- Protocol hierarchy analysis
- PCAP generation
- Basic network reconnaissance concepts

Security Perspective

This lab demonstrated how reconnaissance traffic appears on a network and how packet analysis tools can be used to identify abnormal scanning behavior.

The project helped in understanding:

- how network scans generate identifiable packet patterns
- how defenders can inspect suspicious traffic
- how packet-level visibility supports monitoring and detection

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Conclusion

This isolated virtual lab successfully demonstrated TCP SYN scan behavior and packet-level analysis using Wireshark and TShark.

The project strengthened practical understanding of:

- TCP/IP communication
- Port scanning behavior
- Network traffic monitoring
- Packet analysis workflows
- Security-focused protocol inspection

Disclaimer

All activities were performed strictly inside isolated virtual lab environments for educational and defensive cybersecurity learning purposes only.